

Sampling Distribution

In theory of sampling, we investigate what would be the properties of an arbitrary sample drawn from the population, rather than studying a single sample.

Consider a population of objects, each of which has a numerical value associated with it.

So each observation in the population can be regarded as a value of a random variable having some probability distribution $f(x)$.

If we fix the size of the sample to be 'n', then numerous samples of size 'n' could be drawn from the population.

However, before a sample is chosen, there is uncertainty about the values of the observations. So choosing a sample can be regarded as a random experiment and each numerical observation in a sample as a random variable.

Thus, a random sample of size 'n' from the population can be represented as a set of 'n' random variables.

Let us select a sample of size 'n'; $S = \{x_1, x_2, \dots, x_n\}$.
Once a sample is selected, each of these random variables takes specific values.

For example, ~~let~~ suppose that our population consists of the weights of all students in a college. So a sample of size 5, from this population, before it is selected, can be represented by the set $S = \{x_1, x_2, x_3, x_4, x_5\}$.
When an actual sample of size 5 is picked up, it would represent one set of possible values of the random variables x_1, x_2, x_3, x_4, x_5 .

Suppose that two samples of size 5 chosen from the population turn out to be

$$S_1 = \{52.5, 66.4, 65.6, 56.6, 62.2\}$$

$$S_2 = \{49.5, 58.8, 63.8, 56.1, 60.1\}$$

This means, the random variable X_1 takes value 52.5 in sample S_1 and 49.5 in sample S_2 , but it could take any value from the population data set, similarly for the other random variables.

The value of X_1 varies from sample to sample, but since it could be any value from the original population, X_1 has the same probability distribution $f(x)$ as the original population, the distribution of weights of students in this case.

The same could be the case for any other X_i . So all of them have the same distribution $f(x)$ of the population data.

Moreover, the random variables in S must all be independent, for otherwise it cannot represent a random sample.

Mathematical description of a random sample.

Let $X_1, X_2, \dots, X_n$ represent a random sample of size 'n' from a population. Then they are independent and identically distributed random variables, all having the same probability distribution as the original population.

Sampling distribution of

Consider a population of objects, each of which has a numerical value associated with it. Suppose that the population values of have probability distribution with mean μ and variance σ^2 .

The quantities μ and σ^2 are called population mean and population variance. and in general, such quantities relating the population are called population parameters.

A quantity that can be computed from a sample is called a Statistic.

Since, a sample is a collection of random variables, the statistic is also a random variable. The probability distribution of a statistic is called its sampling distribution.

Sampling distribution of mean

Consider a random sample $X_1, X_2, \dots, X_n$ of size 'n' from a population. The sample mean is a statistic defined as:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

Since, each X_i is a random variable; $\bar{X}$ is also a random variable and its value depends on the sample chosen.

* Since, $\bar{X}$ is a random variable, we can consider the sampling distribution of $\bar{X}$ as the distribution of $\bar{X}$ as the sample size increases.

However, even without knowing the sampling distribution of $\bar{X}$; we can calculate the expected value and variance of $\bar{X}$ as follows

Mean and Variance of Sample mean

Consider a population with population mean μ and variance σ^2 . Let $\bar{X}$ be the mean of a random sample of size 'n' drawn from the population

$$\text{Then, } \bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

$$\begin{aligned} \text{Mean of } \bar{X} &= E(\bar{X}) = E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) \\ &= \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] \\ &= \frac{1}{n} [\mu + \mu + \dots + \mu] \\ &= \frac{1}{n} \cdot n\mu = \mu \end{aligned}$$

$$\begin{aligned} \text{Variance of } \bar{X} &= \text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) \\ &= \frac{1}{n^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] \\ &= \frac{1}{n^2} (\sigma^2 + \sigma^2 + \dots + \sigma^2) \\ &= \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n} \end{aligned}$$

* Thus, the expected value of the sample mean is the population mean μ , and while its variance is $\frac{1}{n}$ times the population variance.

Thus the distribution of $\bar{X}$ is centered at the population mean and its spread is reduced as the sample size 'n' becomes very large

Sampling distribution of mean

Thm.

Let $x_1, x_2, \dots, x_n$ be a sample of size 'n' from a population with mean μ and variance σ^2 and let $\bar{X}$ denote the sample mean.

(1) If the sample size is 'n' is very large ($n \geq 30$), $\bar{X}$ will follow approximately a normal distribution with mean μ and variance σ^2/n .

(2) If the population has a normal distribution, $\bar{X}$ will be exactly normally distributed with mean μ and variance σ^2/n .

Explanation of (2) $\rightarrow$ [If each $X_i \sim N(\mu, \sigma^2)$ then $\bar{X}$, being a linear combination of normal r.v.s, is also a normal rv with mean μ and variance σ^2/n .

$$\therefore, \bar{X} \sim N(\mu, \sigma^2/n)$$

Example:

The weights of students of a graduate college are distributed with mean 63 kg and standard deviation 8.2 kg.

(a) 40 students are chosen at random from the college. What is the probability that their mean weight is less than 65 kg?

(b) The weights of the girls in the college are known to be normally distributed with mean 55 kg and standard deviation 2.2 kg. What is the probability that 9 girls chosen at random from the college will have mean weight between 53 kg and 57 kg?

Answer: (a) Here the population consists of the weights of all students in the college. The population distribution is unknown, but the population mean $\mu = 63$ and standard deviation $\sigma = 8.2$ is given.

Let $\bar{X}$ be the sample mean of an arbitrary sample of size $n=40$ (large sample)

Then $\bar{X}$ has approximately a normal distribution with mean $\mu=63$ and standard deviation

$$\sigma/\sqrt{n} = 8.2/\sqrt{40} = 1.2965$$

$$\text{Hence, } Z = \frac{\bar{X} - 63}{1.2965} \sim N(0, 1)$$

Required probability that the mean weight is less than 65 is

$$P(\bar{X} < 65) = P\left(Z < \frac{65 - 63}{1.2965}\right)$$

$$= P(Z < 1.5426)$$

$$= 0.5 + P(0 < Z < 1.5426)$$

$$= 0.5 + 0.4382 = 0.9382$$

(6) The population here is the weight of girl students in the College, which follow normal distribution with mean $\mu=55$ and standard deviation $\sigma=2.2$

Here the sample size is small ($n=9$), but the population follow normal distribution; So $\bar{X}$ also follow normal distribution with mean $\mu=55$ and standard deviation $\sigma/\sqrt{n} = \frac{2.2}{3} = 0.7333$ for any value 'n'.

$$\text{Then, } Z = \frac{\bar{X} - 55}{0.7333} \sim N(0, 1)$$

$$\text{Now } P(53 < \bar{X} < 57) = P\left(\frac{53-55}{0.7333} < Z < \frac{57-55}{0.7333}\right)$$

$$= P(-2.7274 < Z < 2.7274)$$

$$= 2 \times P(0 < Z < 2.7274)$$

$$= 2 \times 0.4968 = \underline{\underline{0.9936}}$$

Sampling distribution of proportion

In ~~some~~^{many} situations, an investigator wants to determine the proportion of a population having a specific attribute.

For example; the proportion teenagers in a country who are overweight ($BMI > 25$), proportion of defective items produced by a manufacturing company etc.

Each object of the population either possesses the attribute or does not possess the attribute.

Such a population is called a binomial population and an object in the population which possesses the specific attribute is called a Success.

Suppose 'p' is the proportion of success in the population, then p is the probability that each object of the population is a success, independently of others.

i.e., p is the probability of success of the population.

Consider a random experiment of drawing a sample of size 'n' from this population and observing which of them are successes. If X denote the number of successes, then X can be regarded as a binomial random variable with parameters n and p, i.e., $X \sim B(n, p)$.

The proportion of successes in the sample is

$$\hat{p} = \frac{X}{n}$$

We call $\hat{p}$ as sample proportion.

Since, X is a random variable, $\hat{p}$ is also a random variable and its distribution

is called sampling distribution of proportion

Since, $X \sim B(n, p)$, we have $E(X) = np$ and $\text{var}(X) = npq$, where $q = 1 - p$.

$$\text{Hence } E[\hat{p}] = E(X/n) = \frac{np}{n} = p$$

$$\text{and } \text{Var}(\hat{p}) = \text{Var}(X/n) = \frac{1}{n^2} \text{Var}(X) = \frac{npq}{n^2} = \frac{pq}{n}$$

* n is large ($n > 20$) and p is not close to either 0 or 1, then X can be approximated by a normal distribution $N(np, npq)$

Hence, under these conditions $\hat{p}$ can be approximated by a normal distribution with mean p and variance pq/n

* [The approximation gets better as p moves further away from 0 or 1; we can have a rule for validity of this approximation as $np \geq 10$ and $nq \geq 10$]

~~Approximate~~

Theorem

Consider a binomial population in which the proportion of successes is p . Let $\hat{p}$ denote the proportion of successes in a random sample of size n drawn from the population. Then if n is large ($np \geq 10$ and $n(1-p) \geq 10$), then $\hat{p}$ has approximately a normal distribution with mean p and variance pq/n , where $q = 1 - p$.

Q. Approximately 60% of graduate students in a city are females. What is the probability that in a sample of 100 graduates from the city, the number of female students is 58 or less?

Soln. The population proportion of female students is $p = 60/100 = 0.6$, so, $q = 1 - p = 0.4$

Let $\hat{p}$ denote the proportion of female students in a random sample of $n = 100$ students.

Then by theorem

$$\hat{p} \sim N(p, \sqrt{pq/n}) \quad \sqrt{pq/n} = 0.0490$$

$$\text{i.e., } \hat{p} \sim N(0.6, 0.0490)$$

This approximation is valid, since $np = 60 > 10$
and $nq = 40 > 10$

$$\text{Then, } Z = \frac{\hat{p} - 0.6}{0.0490} \sim N(0, 1)$$

Required probability is

$$P(\hat{p} \leq 58/100) = P(\hat{p} \leq 0.58)$$

$$= P\left(Z \leq \frac{0.58 - 0.6}{0.0490}\right)$$

$$= P(Z \leq -0.4082)$$

$$= 0.5 - P(0 \leq Z \leq 0.4082)$$

$$= 0.5 - 0.1584$$

$$= 0.3416$$

